7 Cryptography

7.6 RSA Algorithm

Example 7.6.1 Compute the following:

1. $3^9 \mod 15$
2. $2^9 \mod 15$
3. $9^9 \mod 15$
4. $14^9 \mod 15$

Example 7.6.2 What might we do with the above observation?

Factor 9 into 3 and 3, then notice that this gives rise to a 2-step process, we could compute $b = a^3 \mod 15$, then $c = b^3 \mod 15$, but then $c = a^9 \mod 15 = a \mod 15$. So we can recover what a was.

Definition 7.6.3 (RSA Algorithm) Each party wishing to communicate does the following:

1. Pick two prime numbers, p and q.
2. Compute $n = p * q$.
3. Compute $\phi(n) = (p-1) * (q-1)$
4. Pick a public key e satisfying $\gcd(e, \phi(n)) = 1$.
5. Find d such that $d * e \equiv 1 \mod \phi(n)$.
6. Publish e and n as public keys.
7. Guard p, q, and $\phi(n)$ as secret, and keeps d secret as a secret key.

Question: What makes this secure? What step is hard to compute without the secret keys?

Example 7.6.4 Encrypt the word "help" using the RSA algorithm $n = 3551$, $e = 629$, and $d = 1997$.

Ok... no more hand calculations for this lecture! Current best practices recommend a key length of at least 2048 bits, that means that n should be a number with around 617 digits! Let's see a few examples with a computer.

Go to website, and see some examples of how this looks with larger primes.